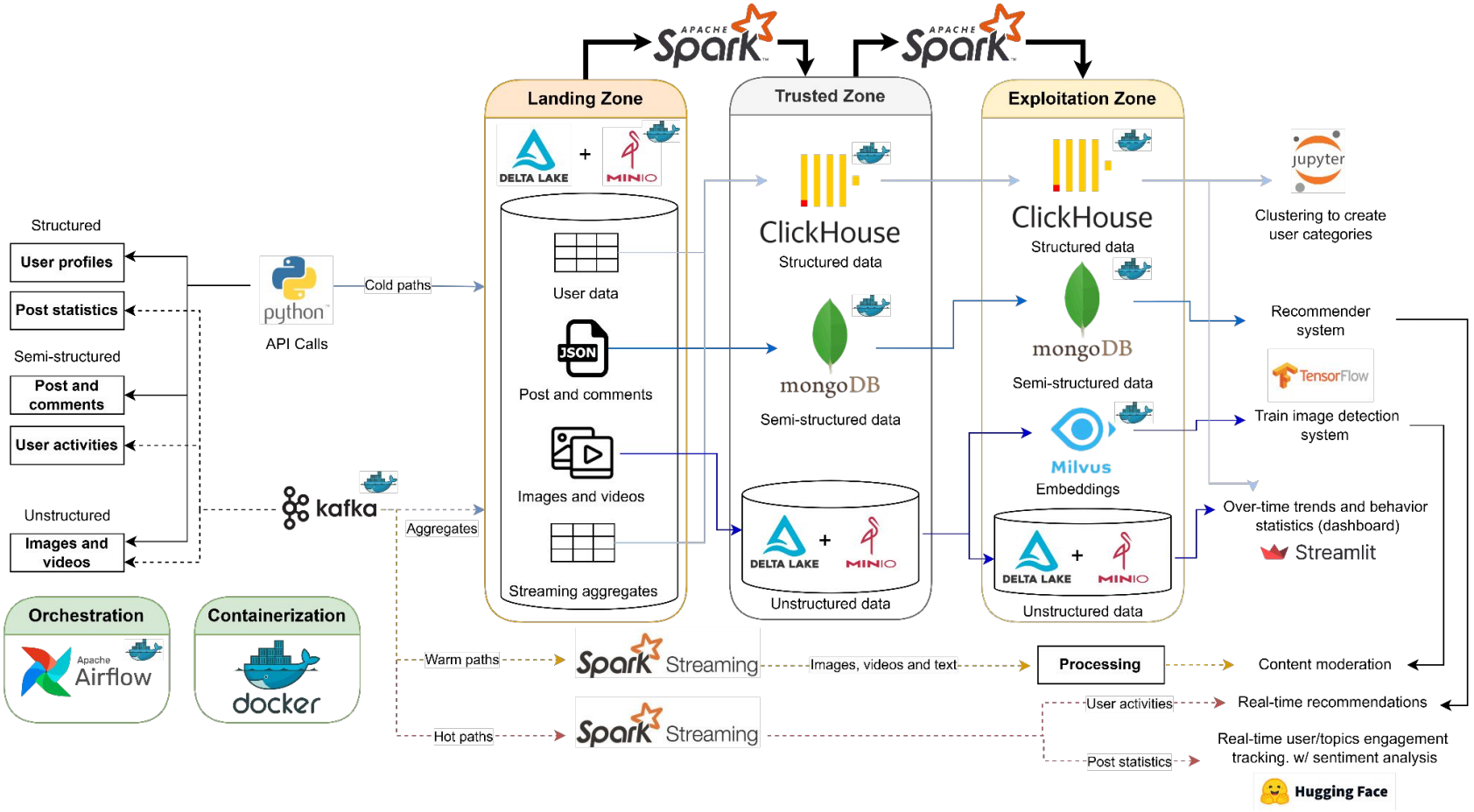
Big Data Management
Project P2

DTIM Research Group
Universitat Politècnica de Catalunya (UPC) - BarcelonaTech

Context	Data Ingestion	Landing Zone	Trusted Zone	Exploitation Zone	Data Consumption
Define domain, value and datasets	Get data from the **sources**	Store the **raw** data	Apply **generic** data quality	**Arrange** data to be exploited	Get the data and generate **value**


Spark
Spark
Landing Zone
Trusted Zone
Exploitation Zone
DELTA LAKE
MINIO
User data
JSON
Post and comments
Images and videos
Streaming aggregates
ClickHouse
Structured data
mongoDB
Semi-structured data
DELTA LAKE
MINIO
Unstructured data
ClickHouse
Structured data
mongoDB
Semi-structured data
Milvus
Embeddings
DELTA LAKE
MINIO
Unstructured data
jupyter
Clustering to create user categories
Recommender system
TensorFlow
Train image detection system
Over-time trends and behavior statistics (dashboard)
Streamlit
Structured
User profiles
Post statistics
Semi-structured
Post and comments
User activities
Unstructured
Images and videos
python
API Calls
Cold paths
kafka
Aggregates
Orchestration
Apache Airflow
Containerization
docker
Warm paths
Spark Streaming
Images, videos and text
Processing
Content moderation
Hot paths
Spark Streaming
User activities
Real-time recommendations
Post statistics
Real-time user/topics engagement tracking. w/ sentiment analysis
Hugging Face

Stages Overview

Trusted Zone

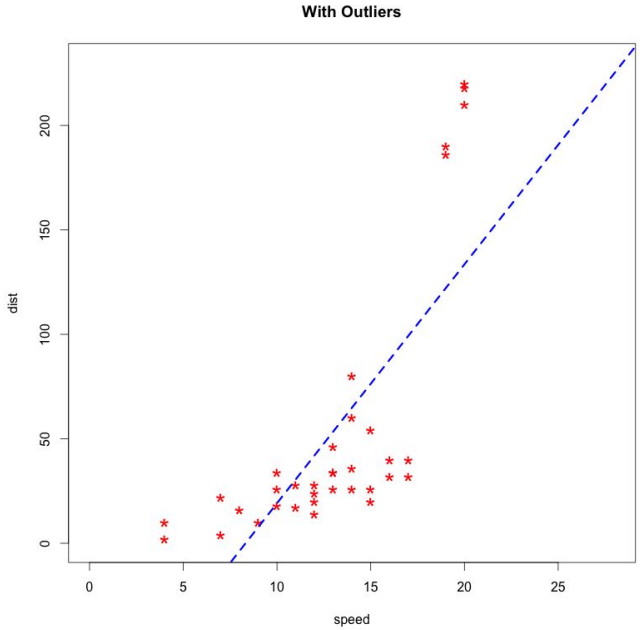
- Raw data (stored in the landing zone) comes with all sorts of mistakes or inconsistencies that need to be address **before** it is used.

Trusted Zone

- Raw data (stored in the landing zone) comes with all sorts of mistakes or inconsistencies that need to be address **before** it is used.

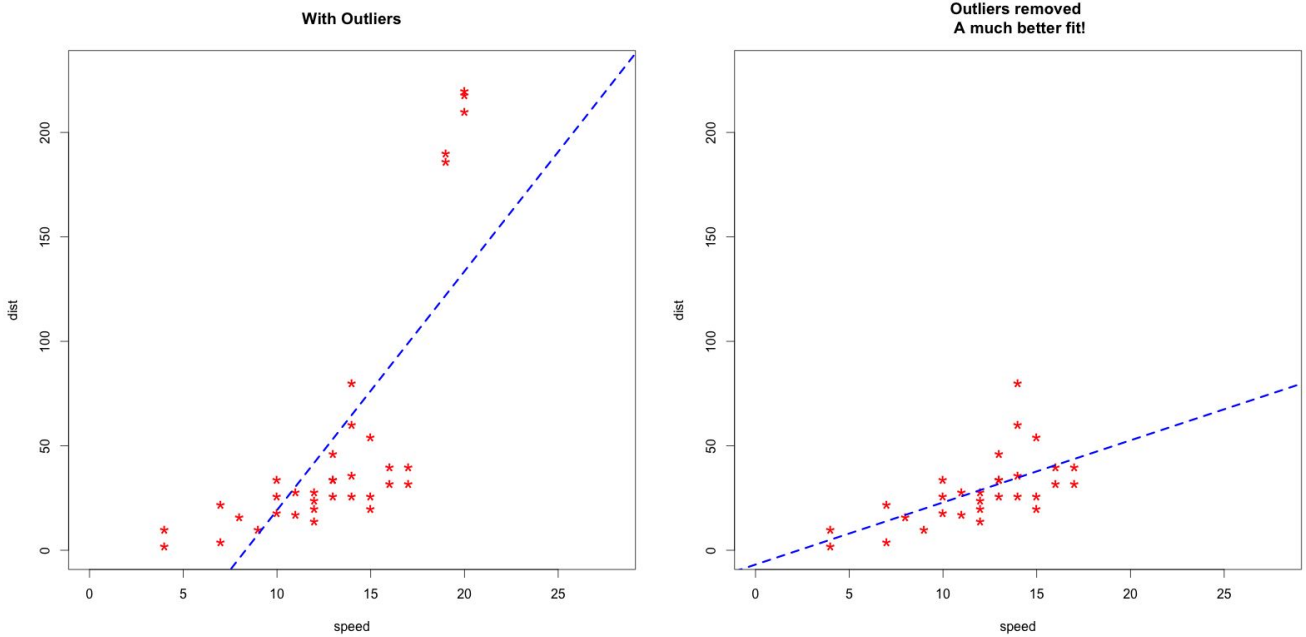
- Our data might be used for a lot of tasks. Hence, we **can not** apply **specific** preprocessing techniques.

Trusted Zone



r-statistics: *Outlier Treatment With R*

Trusted Zone



r-statistics: *Outlier Treatment With R*

Trusted Zone

- Raw data (stored in the landing zone) comes with all sorts of mistakes or inconsistencies that need to be address **before** it is used.

- Our data might be used for a lot of tasks. Hence, we **can not** apply **specific** preprocessing techniques.
  - Be careful with any processing that assume a given downstream task: removing outliers, normalizing data, aggressive text normalization, image alterations, etc.

Trusted Zone

- Raw data (stored in the landing zone) comes with all sorts of mistakes or inconsistencies that need to be address **before** it is used.

- Our data might be used for a lot of tasks. Hence, we **can not** apply **specific** preprocessing techniques.
  - Be careful with any processing that assume a given downstream task: removing outliers, normalizing data, aggressive text normalization, image alterations, etc.

- What we have to apply are **generic** data quality tasks.
  - Our goal is to "fix mistakes".
  - *If two downstream tasks with different requirements consumed this data, would both benefit from this transformation, or would one of them be harmed?*

Trusted Zone

Examples of tasks (more are listed in the document).

- **Structured data**:
  - Deduplication: removing repeated elements (e.g. based on primary key)
  - Schema correction: ensuring data is casted to its proper type.
- **Semi-structured data**:
  - Convert deeply nested objects into a more accessible tabular format
  - Handling missing or optional fields (e.g. add default values).
  - Ensure the data conforms to expected structural rules (e.g. valid JSON schema).
- **Unstructured data**:
  - Text: convert to lowercase, remove extra whitespace, standardize encoding, etc.
  - Images: resolution and size normalization, color / brightness correction, etc.

Trusted Zone - Tasks

Constraint 1: Trusted Zone design. First, select the required data storage tools (see section 2) to implement the Trusted Zone. Second, define where each of your data assets will be stored and which transformations will be applied to each to clean it.

Constraint 2: Trusted Zone implementation. Set up the required tools and implement the pipelines that move the data from the Landing Zone to the Trusted Zone. Do so with the appropriate technology (see section 2).

Factors that will **positively contribute** towards the grade:
- Thoroughness and correctness of the cleaning tasks.

Exploitation Zone

- Data in the trusted zone is clean, but also **poorly organized**. The goal of the exploitation zone is to **provide data in the best possible way to facilitate analyses**.

Exploitation Zone

- Data in the trusted zone is clean, but also **poorly organized**. The goal of the exploitation zone is to **provide data in the best possible way to facilitate analyses**.

- We are going to implement two separate processes to do so:
  - Organizing the data (including both storage and modeling)
  - Creating new asets

Exploitation Zone - Organizing

- Unstructured data **has no schema**, so all we can do is to **store it in folders** that clearly indicate the data that is contained.
  - The same structure employed in the Landing Zone might suffice, or some, more specific, buckets might need to be created.

Exploitation Zone - Organizing

- The same thing applies to structured / semi-structured data, although here we can also **model the data**. That can mean performing:
  - Table joins

Country	GDP	HDI

Country	Military	Land

Country	GDP	HDI	Military	Land

Exploitation Zone - Organizing

- The same thing applies to structured / semi-structured data, although here we can also **model the data**. That can mean performing:
  - Table joins
  - Vertical partitioning

Country	GDP	HDI	Military	*Long-text variable*

Country	GDP	HDI	Military

Country	*Long-text variable*

Exploitation Zone - Organizing

- The same thing applies to structured / semi-structured data, although here we can also **model the data**. That can mean performing:
  - Table joins
  - Vertical partitioning
  - Table union

Day	People	Amount
01-01-2026	22.532	892.133
02-01-2026	24.231	851.514

Day	People	Amount
03-01-2026	20.765	901.512
04-01-2026	22.189	902.319

Day	People	Amount
01-01-2026	22.532	892.133
02-01-2026	24.231	851.514
03-01-2026	20.765	901.512
04-01-2026	22.189	902.319

Exploitation Zone - Organizing

- The same thing applies to structured / semi-structured data, although here we can also **model the data**. That can mean performing:
  - Table joins
  - Vertical partitioning
  - Table union
  - Horizontal partitioning

A	B	C
1	B1	...
2	B1	...
3	B2	...
4	B2	...

A	B	C	D
1	B1	...	...
2	B1	...	...

A	B	C	D
3	B2	...	...
4	B2	...	...

Exploitation Zone - Organizing

- The same thing applies to structured / semi-structured data, although here we can also **model the data**. That can mean performing:
  - Table joins
  - Vertical partitioning
  - Table union
  - Horizontal partitioning

- With semi-structured data the process is more complex due to the flexible and potentially nested structure of the data. Nonetheless some level of transformation and organization is possible.
  - For instance, fields that appear consistently across entries can be extracted and normalized into structured tables.

Exploitation Zone - New Data

- For unstructured data, this consists on generating **new information or artifacts** that enrich the data.
  - **Embeddings**
  - Summaries of textual data.
  - Classify images into categories
  - …

Exploitation Zone - New Data

- For unstructured data, this consists on generating **new information or artifacts** that enrich the data.
  - **Embeddings**
  - Summaries of textual data.
  - Classify images into categories
  - …

- For structured and semi-structured data, **KPIs** or other business-related **aggregates** can be obtained. For example:
  - Total benefit per month.
  - Percentage of returning customers.
  - Average time from job posting to hire.

Exploitation Zone - New Data

Constraint 3: Exploitation Zone design. First, select the required data storage tools (see section 2) to implement the Exploitation Zone. Second, define how the data is going to be organized, both in terms of its storage (e.g. databases, buckets) as well as its modelling (e.g. should we perform joins? Should we split some data?). Third, decide which additional data assets you are going to generate (e.g. KPIs, embedding, labels, etc.) and how.

Constraint 4: Exploitation Zone implementation. Set up the required tools and implement the pipelines that move the data from the Trusted Zone to the Exploitation Zone. Do so with the appropriate technology (see section 2).

Factors that will **positively contribute** towards the grade:
- Effort in providing a coherent and nuanced organization to the data.
- Appropriate development of new data/artifacts

Exploitation Zone - New Data

Scope of the Exploitation Zone.: The amount of restructuring and new assets you can introduce depends on your data and use case; we do not expect extensive transformations or large catalogs of derived artifacts. What we evaluate is whether the organization you propose is coherent with the consumption tasks you have defined and whether the design decisions are well justified.

Data Consumption

- At this point most of the data management aspects have been fulfilled, and **data is ready to be employed** in subsequent tasks or fed to external systems.

Data Consumption

- At this point most of the data management aspects have been fulfilled, and **data is ready to be employed** in subsequent tasks or fed to external systems.

- **Priorities** are:
  - Think about how to exploit your data.
  - Think about which tools to do so (and set them up).
  - Establish the necessary connections to ship the data.

Data Consumption

- At this point most of the data management aspects have been fulfilled, and **data is ready to be employed** in subsequent tasks or fed to external systems.

- **Priorities** are:
  - Think about how to exploit your data.
  - Think about which tools to do so (and set them up).
  - Establish the necessary connections to ship the data.

- Recall that <u>BDM is not a modeling/analysis course.</u>

Data Consumption

- At this point most of the data management aspects have been fulfilled, and **data is ready to be employed** in subsequent tasks or fed to external systems.

- **Priorities** are:
  - Think about how to exploit your data.
  - Think about which tools to do so (and set them up).
  - Establish the necessary connections to ship the data.

- Recall that <u>BDM is not a modeling/analysis course.</u>

- If needed, you can use **placeholders** and / or **synthetic data**.

Data Exploitation- Tasks

Constraint 5: Data consumption design and implementation First, define which mechanisms you are going to use to exploit the data. Then, select the tools you will use to implement such mechanisms and set them up. Finally, ship the data from the Exploitation Zone onto selected systems.

Factors that will **positively contribute** towards the grade:
- Setting up minimal, although functional, tasks rather than placeholders.

Implementing the zones

Design

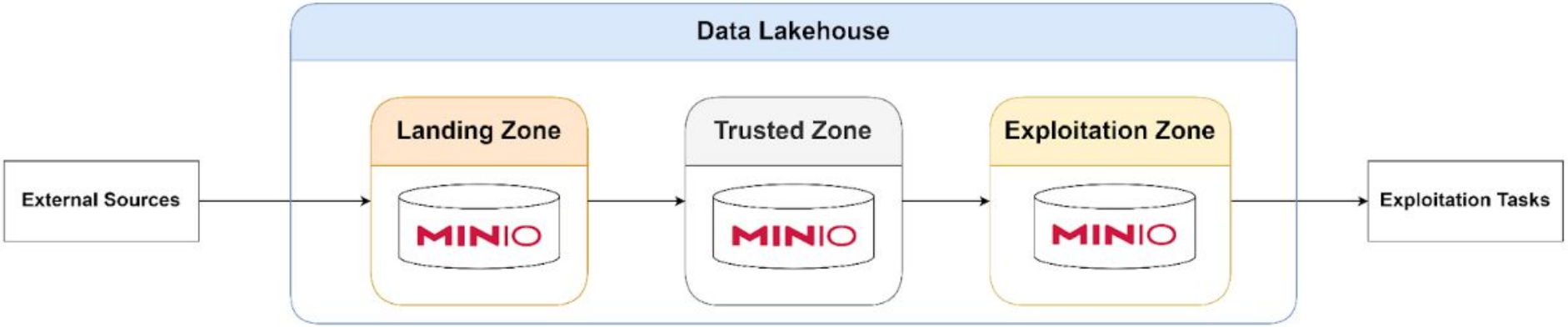
- The most important part is that **each zone has to store the data** after the respective transformations have been applied.

Design

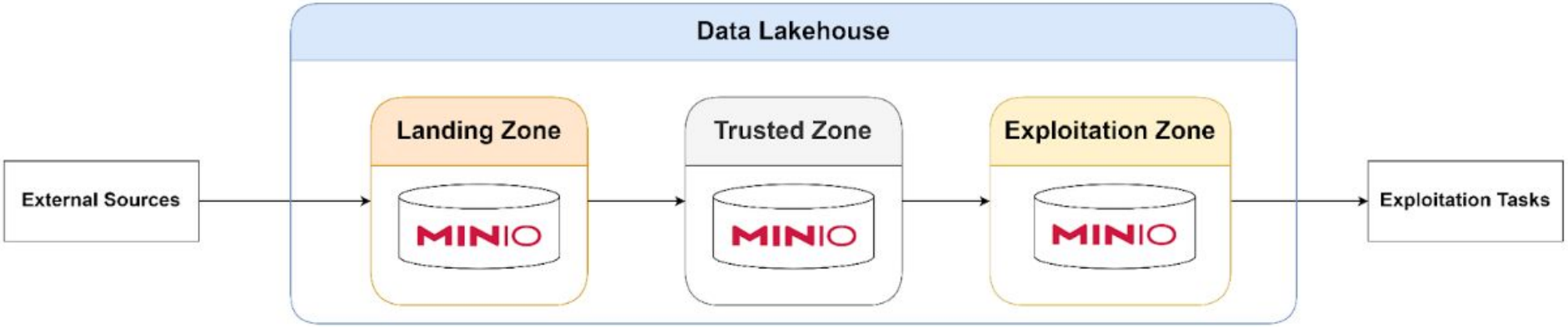
- The most important part is that **each zone has to store the data** after the respective transformations have been applied.

- We propose two main design alternatives

Design - Option 1: Extending your Implementation

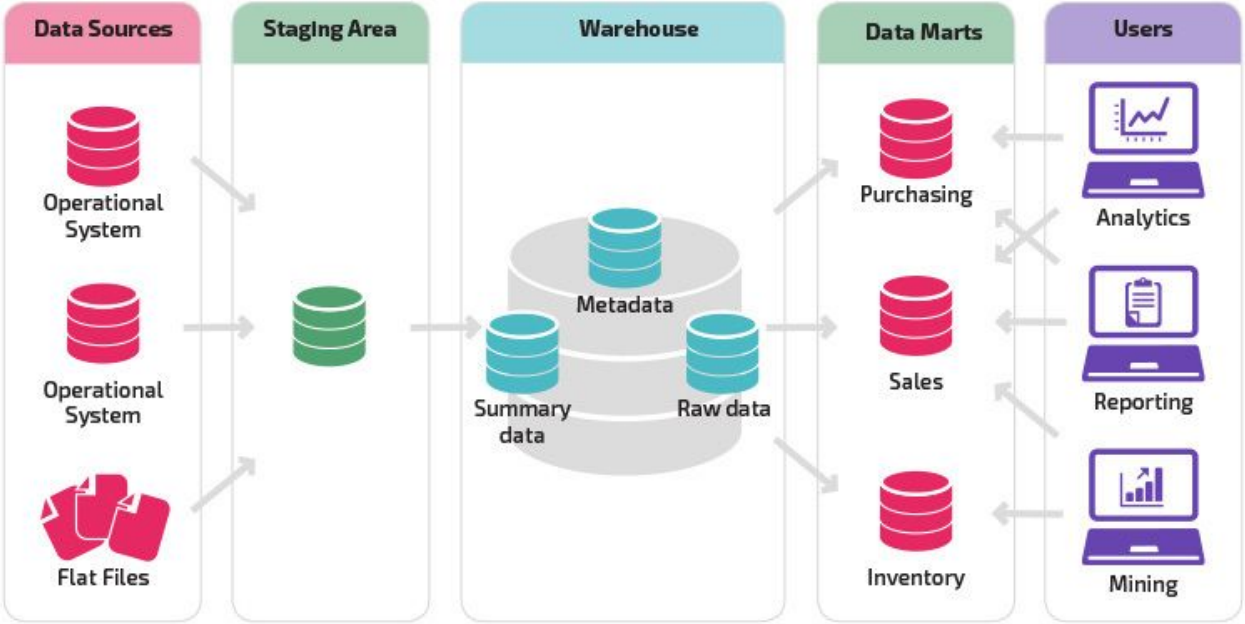


Design - Option 1: Extending your Implementation



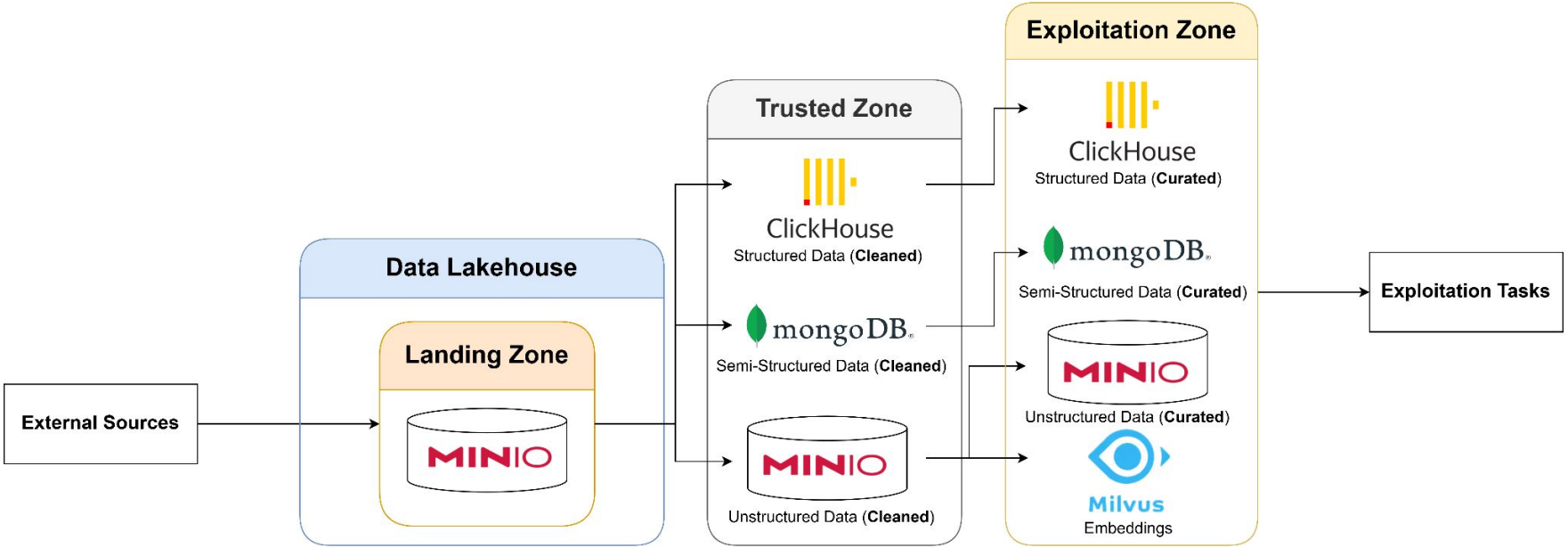
Most common approach in real data management systems

Design - Option 2: Moving the Data Out



Panoply: *Data Mart vs. Data Warehouse*

Design - Option 2: Moving the Data Out



Design - Tasks

Constraint 6: Architecture design. Continuing with the work you did with part 1, you have to complete the architecture diagram with the new zones, tools, operations, etc. Be sure to adequately reason about the general structure you propose and justify the inclusion of each element.

Regarding the presented design alternatives, option 1 is more straightforward, whereas option 2 can allow you to make a more interesting pipeline. From an evaluation standpoint, both are perfectly acceptable. Also note that you can find an **example of the full project diagram** at the end of this document.

Databases

Databases

- Relational Databases
  - Transactional (“traditional”): ORACLE MySQL PostgreSQL
  - Analytical (“modern”): DuckDB ClickHouse

Databases

- Relational Databases
  - Transactional (“traditional”): ORACLE MySQL PostgreSQL
  - Analytical (“modern”): DuckDB ClickHouse
- Key-Value Stores redis

Databases

- Relational Databases
  - Transactional (“traditional”): ORACLE MySQL PostgreSQL
  - Analytical (“modern”): DuckDB ClickHouse
- Key-Value Stores redis
- Document Stores mongoDB

Databases

- Relational Databases
  - Transactional (“traditional”): ORACLE MySQL PostgreSQL
  - Analytical (“modern”): DuckDB ClickHouse
- Key-Value Stores redis
- Document Stores mongoDB
- Wide-Column APACHE HBASE Apache CASSANDRA

Databases

- Graph Databases

Databases

- Graph Databases
- Time-Series Databases

Databases

- Graph Databases
- Time-Series Databases
- Vector Databases

Transformations

Transformations

- Batch

Transformations

- Batch

- Streaming:
  - Should your process the data you stream?
    - *Conceptually*, not in a hot path, as data has to be sent immediately.
    - *Conceptually*, yes in a warm path, as we can tolerate a higher latency.

Transformations

- Batch

- Streaming:
  - Should your process the data you stream?
    - *Conceptually*, not in a hot path, as data has to be sent immediately.
    - *Conceptually*, yes in a warm path, as we can tolerate a higher latency.
  - Tools to process a stream

(In Java)

Data Governance (Optional)

Data Governance

Some aspects of the project, such as tools like OTFs and tasks like metadata handling, are related to **data governance**. In this section we will present a more in-depth exploration of what data governance entails, given that it plays a pivotal role in large-scale data management. Note, however, that the governance tasks we describe below **are not mandatory**, and including governance processes will simply **positively contribute** towards the grade. Please check the guidelines at the end of the section.

Deliverables

Main deliverable

- It has to implement:
  - **Architecture design** (updated version with the new zones, tools, processes, etc.)
  - Stage 3: trusted zone
  - Stage 4: exploitation zone
  - Stage 5: consumption tasks
  - Governance tasks (if applicable)

Main deliverable

Constraint 7: orchestration and containerization (continued from P1). The orchestration and containerization set up in P1 must be extended to cover the new zones and processes introduced in P2. That is, the end-to-end pipeline (ingestion → Landing → Trusted → Exploitation → Consumption) should be orchestrated and, where services are deployed, containerized. The evaluation criteria established in P1 (Constraints 8 and 9) continue to apply. As in P1, additional aspects such as error handling and reporting, execution monitoring or code quality tooling will positively contribute towards the grade.

Constraint 8. Organization. Clearly organize your code and storage repositories per zones / stages. Each zone should have a folder with its respective scripts. Clearly separate the scripts per task (if / when required).

Follow-up deliverables

- First deliverable:
  - Architecture design (first version).

- Second deliverable:
  - Architecture design (updated).
  - Basic implementation of trusted zone, exploitation zone and consumption tasks.